

200 Gbps Coherent Optical Switching Enabled by an Integrated Si₃N₄-TFLN Laser

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Abstract We achieve 200 Gbps coherent optical switching with a switching latency of 80 ns and bit error ratios of below 1×10^{-4} . It is enabled by a tri-material Integrated Si₃N₄-TFLN laser unifying an intrinsic linewidth of 181 Hz and below-15-ns electro-optic wavelength switching. ©2026 The Author(s)

Introduction

The explosive growth of data center traffic driven by cloud computing and AI imposes stringent demands on the low-latency switching networks. Optical circuit switching (OCS) has emerged as a promising solution to replace conventional electronic switching, benefiting from its flattened network topology and high scalability [1]. Among various OCS architectures, wavelength routing scheme notably excels owing to its ns-level switching latency and passive network core [2]. Compared to current intensity modulation and direct detection (IM-DD) format in OCS, coherent technology offers ultra-high throughput and strong immunity to insertion loss. Hence, it is gradually migrating toward short-reach interconnect and switching scenarios in recent years [3,4].

In coherent switching systems, fast-tunable chip-scale lasers play a central role in determining overall system performance. In particular, the tuning speed governs the switching latency, while the laser linewidth fundamentally limits the achievable data rate. Conventional semiconductor lasers [5,6] and recently emerged silicon-based lasers [4,7] typically employ carrier-based fast tuning, but it introduces unwanted intensity modulation and additional noise resulting in degradation of the overall switching performance. With the continuous success in integrated photonics, new integrated platforms for high-speed tunable lasers have emerged. In particular, we highlight thin-film lithium niobate (TFLN)-based integrated lasers, which utilize strong electro-optic (EO) effect for noise-free ultrafast wavelength tuning [8]. In our previous work, we proposed a family of multi-material integrated lasers that unify ultra-low noise and ultrafast EO tunability [9], which hold huge untapped potential towards

low-latency OCS.

In this work, we report a hybrid Integrated Si₃N₄-TFLN laser to achieve coherent optical switching. This laser Si₃N₄-TFLN combines intrinsic linewidth of 181 Hz and below-15-ns EO wavelength switching. It drives a 25 Gbaud dual-polarization 16-state quadrature amplitude modulation (DP-16QAM) switching system, featuring a switching latency of 80 ns (excluding DSP processing time) and bit error ratios remains below 1×10^{-4} .

Device description and characterization

Fig. 2(a) exhibits our demonstrated hybrid integrated Si₃N₄-TFLN laser. Basically, it consists of the tri-material hybrid integration of a C-band InP semiconductor optical amplifier (SOA) and two external cavities on Si₃N₄ and TFLN platforms, respectively. The Si₃N₄ chip is fabricated on a 100-nm-thick wafer-scale platform featuring a loaded quality factor (Q_L) of 0.7 million, while the TFLN chip is fabricated on an X-cut 400-nm-thick platform with a Q_L of 0.08 million. These two external cavities serve as microring-based laser reflectors, with measured butt-coupling losses of 1.8 dB (Si₃N₄-SOA) and 2.7 dB (TFLN-SOA). They present similar free spectral range (FSR) of 51 GHz with slightly difference, enabling a Vernier tunability of 5 THz. By combining ultrafast TFLN photonics and low-loss Si₃N₄ photonics, this laser unifies high coherence and high-speed tunability. Specifically, harnessing the EO dynamics of the TFLN microring facilitates high-speed laser wavelength switching, while the high-Q Si₃N₄ microring underpins a narrow linewidth of the laser.

We first characterize the linewidth performance of the Si₃N₄-TFLN laser. It presents single-mode lasing at 1534.2 nm with an output power of 7.2 mW. Figure 1(b) exhibits the measured single-sided frequency noise

spectrum and its corresponding integral phase noise curve, from which we extract an intrinsic linewidth of 181.4 Hz and an integral linewidth of 7.79 kHz. The achieved narrow linewidth benefits from the high-Q Si_3N_4 microring.

Next, we demonstrate ultrafast EO switching of the laser, as the experimental setup shown in Fig.1(c). We apply a square waveform on TFLN microring to trigger periodic wavelength switching between 1534.2 nm and 1536.6 nm. In order to monitor the switching dynamics, the laser output is split into two branches, where each branch is heterodyned with a reference laser whose wavelength corresponds to a specific switching mode. Figure 1(d) shows the resulting beat-note waveforms in the two branches, confirming successful wavelength

switching. Figure 4(f) provides enlarged views of the beat-note signals around the switching edges, from which the switching times (defined as the interval from >10% power and frequency deviation of the initial mode to <10% deviation of the target mode) are extracted as 13.6 ns and 11.1 ns, respectively. Furthermore, the laser is simultaneously heterodyned with two reference lasers, followed by time–frequency analysis. As shown in Fig. 1(e), the switching behavior is clearly observed. Notably, no measurable frequency drift or power fluctuation is observed between consecutive switching events. The demonstrated high-speed and stable wavelength switching validates the ultrafast EO tunability with negligible noise of the Si_3N_4 -TFLN laser.

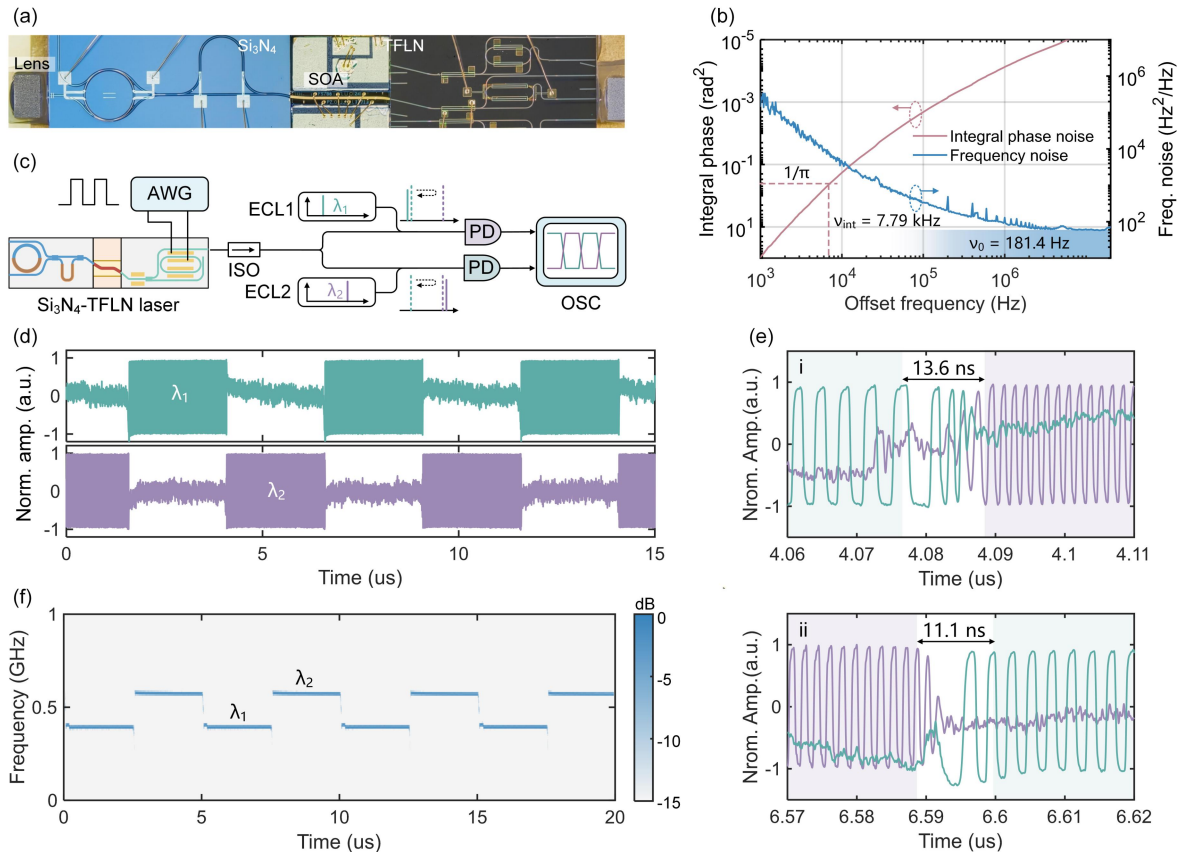


Fig. 1 Fast tunable hybrid integrated Si_3N_4 -TFLN laser. (a) Package image of the demonstrated hybrid integrated Si_3N_4 -TFLN laser. (b) Frequency noise spectrum and the corresponding integral phase noise at 1539.05 nm. (c) Experimental setup for wavelength switching. AWG, arbitrary waveform generator; ISO, isolator; ECL, external cavity laser; PD, photodetector; OSC, oscilloscope. (d) Switching waveforms of the two wavelength channels with a spacing of 2.4 nm. (e) Enlarged views in (f), yielding switching times of below 15 ns. (f) Time-frequency spectrum of beat note between the laser with two reference lasers, indicating the entire switching dynamics.

Coherent Switching Experiment

By unifying narrow linewidth and ultrafast EO tunability, the Si_3N_4 -TFLN laser empowers high-throughput, low-latency coherent OCS networks. We carry out a proof-of-concept experiment to validate its performance using a 25 Gaud DP-16QAM system, with the setup shown in Fig. 2(a). The Si_3N_4 -TFLN laser operates at a wavelength-switching state, and two reference

lasers are combined and used as local oscillators for time-interleaved coherent detection corresponding to the two switching modes. The received signal, which contains the switching process and transmission data at two switching wavelengths, is recorded by a real-time oscilloscope and processed offline. The digital signal processing (DSP) flow includes IQ orthogonalization, clock recovery, multi-input-multi-output equalization, frequency offset

estimation, carrier phase recovery, decision-directed least mean squares equalization and final decision.

Figure 2(b) exhibits the time-varying frequency offset information extracted from the received signal, where the observed frequency hopping indicates successful wavelength switching. To quantify the switching latency, a sliding time window of 1.6 μs centred around the switching trigger edge is employed for data recovery. Figure 2(c) shows the signal-to-noise ratio (SNR) of the recovered constellation diagram as a function of the sliding window position. The initial decrease followed by a subsequent increase in SNR reflects the transition of the sliding window from one wavelength into the switching region and then into the target wavelength. The switching latency (excluding DSP processing time) is defined as the duration of the switching region with invalid data and is estimated to be 80 ns. It should be noted that this value is influenced by

the employed DSP algorithms. With DSP schemes specifically optimized for coherent switching architectures and featuring faster convergence, a more accurate and potentially shorter switching latency could be achieved [4]. In fig. 2(d), we recover the constellation diagrams of the signals outside the switching region at the two wavelengths. The calculated bit error ratios of 7.5×10^{-5} and 2.5×10^{-5} are well below the 7% hard-decision forward error correction (HD-FEC) threshold of 3.8×10^{-3} . Notably, the line rate of 200 Gbps is primarily limited by the operational bandwidth of our employed transmitter and receiver. Looking ahead, we anticipate that the Si_3N_4 -TFLN laser could further support beyond-400-Gbps coherent switching. These results confirm the practical viability of the Si_3N_4 -TFLN laser for high-speed optical switching in coherent systems.

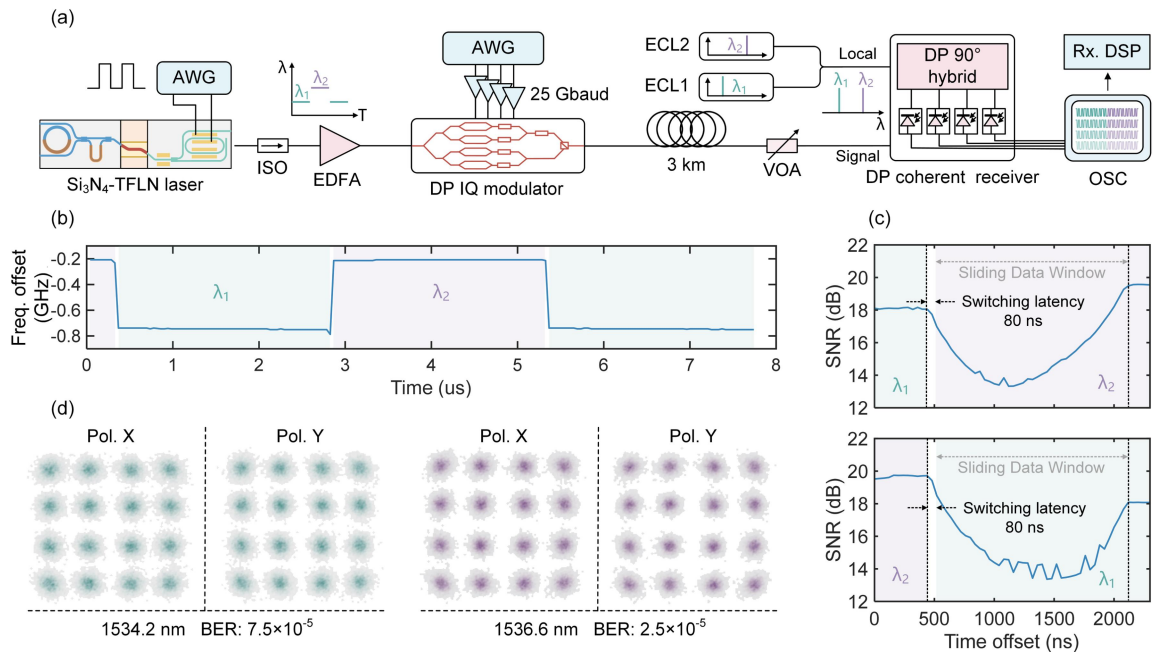


Fig. 2 Optical coherent switching experiment. (a) Architecture of the optical coherent switching system. EDFA, erbium-doped fiber amplifier; DP IQ modulator, dual-polarization in-phase/quadrature modulator; VOA, variable optical attenuator; Rx., transmitter. (b) Time-varying frequency offset information recovered from the received data. (c) SNR of the recovered constellation diagram changing with the sliding time window, indicating the switching latencies of 80 ns. (d) Recovered constellation diagrams at the two switching wavelengths.

Conclusions

In conclusion, we have utilized a hybrid Integrated Si_3N_4 -TFLN laser to achieve 200 Gbps coherent optical switching. By leveraging ultrafast TFLN photonics and low-loss Si_3N_4 photonics, the laser combines an intrinsic linewidth of 181 Hz and below-15-ns EO wavelength switching. It drives a 25 Gbaud DP-16QAM switching system, where we achieve a

switching latency (excluding DSP time) of 80 ns and a the BER remains below 1×10^{-4} . The demonstrated coherent switching open up new opportunities towards low-latency and high-throughput OCS networks.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No.62522502, 62371056).

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